meta mapg

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1 Glossary

Symbol	Meaning
Stochastic game components	
$\mathcal{G}$	Stochastic game tuple
$\mathcal{S}$	Finite set of states
$\mathcal{N}$	Finite set of agents
$\mathcal{A}_i$	Action set of agent i
$\mathcal{A}$	Joint action space
$\mathcal{A}_{-i}$	Action space of all agents except i
R_i	Reward function of agent i, bounded by $R_{\max}$
$P(s' \mid s, \alpha)$	Transition probability
γ_{disc}	Discount factor
ρ	Initial state distribution
Policies and parameters	
$\pi_{\phi_i}(a_i \mid s)$	Policy of agent i
$\phi_i \in R^{d_i}$	Policy parameter of agent i
ϕ_{-i}	Policy parameters of all agents except i
$\phi = (\phi_i)_{i=1}^N \in R^d$	Joint policy parameter
ϕ^*	Nash equilibrium policy
ϕ^*_λ	Shifted fixed point under shaping strength λ
ϕ_0	Initial policy parameter
ϕ_n	Iterate at step n
$\mathcal{K}$	Compact constraint set
Value functions and gradients	
$V_{i,s}(\phi)$	Value function of agent i from state s
$V_{i,\rho}(\phi)$	Value function of agent i under initial state distribution ρ
$G^i(\tau)$	Discounted return of agent i along trajectory τ

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Symbol	**Meaning**
$\tau = (s_t, a_t)_{t \geq 0}$	Trajectory
$v(\phi)$	Policy-gradient update direction
$S_i(\tau)$	Score function $\nabla_{\phi_i} \log p_\phi(\tau)$
$H_i(\tau)$	Hessian $\nabla^2_{\phi_i} \log p_\phi(\tau)$
Nash conditions	
μ	SOS drift constant
ρ	SOS radius
r_{att}	Attraction radius
$B_{r_{\text{att}}}(\phi^*)$	Attraction region of ϕ^*
Meta-MAPG update	
α^i	Learning rate of agent i
α^{-i}	Learning rates of peer agents
L	Unroll length
ℓ	Inner-loop index
$M_L(\phi)$	Meta-MAPG correction
$F_L(\phi)$	Full Meta-MAPG update $v(\phi) + M_L(\phi)$
λ_n	Shaping coefficient schedule
λ_c	Constant shaping coefficient in phase 1
α_n	Outer step size schedule
$\widehat{M}_{L_n,n}$	Sampled Meta-MAPG correction
$\hat{v}_n$	Sampled policy-gradient estimate
Stochastic approximation	
F	Effective update direction
ξ_{n+1}	Martingale-difference noise
σ^2	Variance bound on ξ_{n+1}
b_n	Bias term
β_n	Bias magnitude sequence
C	Bias constant
$\mathcal{F}_n$	Filtration at step n
L_n	Unroll length at step n
Basin entry	
T	Warm-up horizon
$p_{\text{entry}}(\mathcal{A}, T)$	Basin-entry probability of algorithm $\mathcal{A}$
μ_M	Drift improvement from Meta-MAPG correction
ρ_λ	Certified SOS radius under shaping strength λ